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MILITARY SPECIFICATION

BALLOONS, AEROLOGICAL

(This specification has been approved by the Bureau of Aeronautics, Department of the Navy)

1. SCOPE AND CLASSIFICATION

1.1 Scope.-This specification covers free-flight, single envelope, freely expanding balloons made from synthetic rubber latex.

1.2 Grades and Types.-Balloons covered by this specification shall be of one grade and of the following types, classified in accordance with their function, nominal weight, and color as specified in the invitation for bids.

Table I

Types

Function	Nomenclature	Nominal Weight, Grams	Color
Ceiling	Balloon ML-157-A	10	Black or dark blue
Ceiling	Balloon ML-158-A	10	Red
Pilot	Balloon ML-50-A	30	Uncolored
Pilot	Balloon ML-51-A	30	Black or dark blue
Pilot	Balloon ML-64-A	30	Red
Pilot	Balloon ML-155-A	30	Orange
Pilot	Balloon ML-156-A	30	Yellow
Pilot	Balloon ML-159-A	100	Uncolored
Pilot	Balloon ML-160-A	100	Black or dark blue
Pilot	Balloon ML-161-A	100	Red
Pilot	Balloon ML-408A/AM	100	Orange
Pilot	Balloon ML-478/AM	100	Yellow
Sounding	Balloon ML-479/AM	600	Uncolored

2. APPLICABLE SPECIFICATIONS AND STANDARDS

2.1 General.-The following specifications and standards, of the issues in effect on the date of invitation for bids, form a part of this specification.

2.2 Specifications

Federal Specifications

ZZ-R-601 - Rubber Goods; General Specifications (Methods of Physical Tests and Chemical Analyses)
TT-C-595 - Colors (for) ready-mixed paints
UU-P-268 - Paper, Kraft, Wrapping
NN-B-591 - Boxes, Fiberboard, Wood-cleated (for Domestic Shipment)
NN-B-601 - Boxes, Wood-cleated-plywood (for Domestic Shipment)
NN-B-621 - Boxes, Wood, Nailed and Lock-corner
LLL-B-631 - Boxes, Fiber, Corrugated (for Domestic Use)

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Military Specifications

JAN-P-105 - Packaging and Packing for Overseas Shipment-Boxes, Wood, Cleated, Plywood
JAN-P-106 - Packaging and Packing for Overseas Shipment-Boxes, Wood, Nailed
MIL-B-107 - Boxes, Wood, Wirebound (Overseas Type)
JAN-P-108 - Packaging and Packing for Overseas Shipment-Boxes, Fiberboard,(V-Board and W-Board), Exterior and Interior
MIL-P-116 - Preservation, Methods of
JAN-P-120 - Packaging and Packing for Overseas Shipment-Cartons, Folding, Paperboard
JAN-P-127 - Packaging and Packing for Overseas Shipment-Tape, Adhesive, Pressure-Sensitive, Water-Resistant
JAN-P-133 - Packaging and Packing for Overseas Shipment-Boxes, Set-up, Paperboard

Military Standards

MIL-STD-129 - Marking of Shipments

2.3 Availability of Specifications and Publications

2.3.1 When requesting specifications or publications refer to both title and number.

2.3.2 Copies of this specification and applicable specifications, with the exception of Federal Specification TT-C-595, may be obtained upon application to the Commanding Officer, U.S. Naval Air Station, Johnsville, Pennsylvania, Attention: Technical Records Division.

2.3.3 Copies of Federal Specification TT-C-595 may be obtained upon application accompanied by check, money order or cash in the amount of $4.50, to the General Services Administration, Business Service Center, Region 3, Room 7110, Seventh and D Streets, S.W., Washington 25, D. C.

3. REQUIREMENTS

3.1 Bid Samples of Ceiling and Pilot Balloons.- Subject to the bid request, bid samples may be required in the sizes given in Table V. These samples shall be furnished at no cost to the Government, shall be an integral part of the bid, and must be received at or before the time of bid opening. Bid samples may be subjected to any or all of the tests of this specification. The failure of a bid sample to meet the requirements of this specification shall constitute sufficient grounds for the rejection of the bid. Bid samples will not be returned to the bidder whether or not the bid is accepted. Bid samples shall in all respects be representative of the balloons the bidder expects to furnish in production.

3.2 Material. - Balloons furnished under this specification shall be made from a good quality synthetic rubber latex.

3.3 Condition of the Film.- The balloons shall be manufactured and finished in accordance with good commercial practice. The film shall be continuous and shall contain no patches. Unless specifically provided in the contract, this requirement shall not be so construed as to prohibit acceptance of double neck or attached neck balloons which otherwise meet the requirements of this specification. Balloons which have abrasions, pinches, slugs, foreign matter, pinholes, or other defects are not acceptable. Balloons shall feel dry and not greasy, and the balloon film shall be coated inside and outside with talc or equivalent material to prevent the folds from adhering to each other. The balloons shall not be more than 240 days old when submitted for inspection and testing. (See 4.2.)

3.4 Weight. -The weight of a balloon shall be as specified in Table II. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group A given in Table VI. (See 4.3.3.)

3.5 Flaccid Length.-The flaccid length of a balloon shall not be less than the minimum value given in Table II. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group A given in Table VI. (See 4.3.4.)

3.6 Neck.-

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3.6.1 Neck construction. - A tubular neck of approximately constant cross-sectional area shall be molded or dipped integrally with the body of the balloon or securely attached thereto. If the neck wall thickness measured as described in 4.3.5.3 is less than 0.012 inches, the free end of the neck shall have a bead formed around the edge for reinforcement. (See 4.2.)

3.6.2 Neck length. - The length of the neck shall not be less than the minimum value given in Table II. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group A given in Table VI. (See 4.3.5.1.)

3.6.3 Neck diameter. - The inside diameter of the neck shall be as specified in Table II. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group A given in Table VI. (See 4.3.5.2.)

3.6.4 Neck wall thickness. - The neck wall thickness shall be as specified in Table II, when tested in the manner described in 4.3.5.3. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group B given in Table VI.

3.6.5 Neck strength. - The neck of a sounding balloon shall be capable of supporting a weight of 40 pounds for 30 seconds when tested in the manner described in 4.3.5.4. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group B given in Table VI.

3.6.6 Neck marking. - The function and the nominal weight of each balloon, and the date of manufacture of each sounding balloon, shall be stamped or written in indelible ink on the neck of the balloon. (See 4.2.)

Table II

Weight and Dimensions

Function	Total Weight, Grams			Neck Dimensions, inches					Placoid Length inches
				Length	Inside dia.		Wall Thickness		
	Nominal	Min.	Max.	Min.	Min.	Max.	Min.	Max.	Min.
Ceiling	10	8.5	12	2	7/8	1-1/8	0.004	0.020	7-1/2
Pilot	30	26.5	34	2	7/8	1-5/8	0.004	0.040	11
Pilot	100	90	110	2	7/8	1-5/8	0.004	0.060	22
Sounding	600	500	700	4-1/2	7/8	1-5/8	0.020	0.070	72

3.7 Color. - This paragraph does not apply to uncolored balloons. Dyes and pigments used in coloring balloons shall be evenly distributed throughout the material and shall be of sufficient density and brilliance that the balloon will be glossy and nearly opaque and show no streaks when inflated to the diameter or volume given in Table III. When viewed against a black background at the diameter or volume shown in Table III, the color of the balloon shall be similar to the appropriate color given in 3.7.1, 3.7.2, 3.7.3, or 3.7.4. The color numbers in 3.7.1, 3.7.2, 3.7.3, and 3.7.4 refer to Federal Specification TT-C-595. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group B given in Table VI. (See 4.3.6.)

3.7.1 Black. - When black is specified it shall be black No. 1775, dark blue No. 1610, or a similar dark color, so that the balloon will be opaque and glossy when inflated.

3.7.2 Red. - When red is specified it shall be red No. 1106.

3.7.3 Orange. - When orange is specified it shall be orange No. 1210.

3.7.4 Yellow. - When yellow is specified it shall be lemon yellow No. 1320.

3.7.5 Uncolored. - When uncolored balloons are specified no dye or pigment or any other coloring material not required for the compounding or curing of the balloons shall be added.

3.8 Shape. - Ceiling and pilot balloons when tested by the method of 4.3.7 shall have a diameter ratio of at least 90%. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group B given in Table VI.

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Table III

Conditions for Testing Color and Shape

Function	Nominal Weight, grams	Horizontal Diameter, inches	Volume, cubic feet
Ceiling	10	17	1.5
Pilot	30	25	4.8
Pilot	100	39	18.0

3.9 Bursting dimensions. - Ceiling and pilot balloons when tested by the method of 4.3.8 shall have a bursting diameter or volume at least as large as the value given in Table IV. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group B given in Table VI.

Table IV

Bursting Dimensions

Function	Nominal Weight, grams	Horizontal Diameter, inches, min.	Volume, Cubic Feet min.
Ceiling	10	30	8.2
Pilot	30	44	26
Pilot	100	90	221

3.10 Breaking strength and ultimate elongation. - This paragraph does not apply to ceiling balloons. When three (3) dumbbell specimens are cut from each of three balloons and tested in the manner described in 4.3.9, the median breaking strength of the nine (9) specimens shall not be less than 1.00 pound and the median ultimate elongation shall not be less than 700% for sounding balloons and not less than 750% for pilot balloons. A lot shall be rejected when either median is below that specified.

3.11 Differential bursting pressure and effective elongation. - This paragraph applies only to sounding balloons. When three (3) patches are cut from each of three (3) balloons and tested by the method of 4.3.10, the average differential bursting pressure of the nine (9) patches shall not be greater than 110 inches of water and the effective elongation, calculated by the method of 4.3.10.1 shall not be less than 350%. A lot shall be rejected when the samples fail to meet the minimum values of this paragraph.

3.12 Ozone resistance. - This paragraph applies only to sounding balloons. When dumbbell specimens are tested by the method of paragraph 4.3.11 they shall withstand 15 minutes exposure at a temperature of 25 degrees ± 4 degrees centigrade to a concentration of 0.010% to 0.015% ozone without breaking. A lot shall be rejected for failure to meet this requirement in accordance with the plan for test group B given in Table VI.

3.13 Flight elevation. - Sounding balloons shall have a median day flight elevation and a median night flight elevation of at least 23.5 kilometers above mean sea level when tested as described in 4.3.12 and computed in accordance with Table VII. A lot shall be rejected when the median flight elevation, day or night, fails to meet the requirements of this paragraph.

4. SAMPLING, INSPECTION, AND TEST PROCEDURE

4.1 Sampling

4.1.1 Samples. - A sample of the size given in Table V shall be selected at random by a Government inspector from each lot as defined in 4.1.2 and subjected to all of the tests of this specification. All samples selected shall be visually examined by the Government inspector to determine compliance with paragraph 3.3 prior to any further tests required by this specification.

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4.1.2 Lots. - Only balloons of a single function, nominal weight, material and color shall be grouped together into a lot. A lot shall consist of balloons made from a single batch of compounded latex within a period not exceeding seven (7) consecutive calendar days. Each lot shall contain a grouping of balloons that are as nearly homogeneous as possible. The maximum and minimum sizes of production lots are given in Table V. The balloons required for inspection and for shipment to other test facilities as production lot samples are not included in these totals. If the contract calls for a smaller number of balloons then the minimum given in Table V, that number shall be considered a lot and a sample of the size shown in Table V shall be selected. If the number of balloons remaining at the end of a contract is half or more of the minimum lot size, it shall be treated as a lot; if it is less than half of the minimum lot size, it shall be considered as part of the next preceding lot and grouped with it before samples are selected.

Table V

Sample and Lot Sizes

Function	Nominal Weight, grams	Size of Bid Sample	Size of Production Lot Sample	Lot Size min.	Lot Size max.
Ceiling	10	9	6	2000	5100
Pilot	30	9	6	2000	5100
Pilot	100	9	6	2000	5100
Sounding	600		9	500	1000
Sounding	600		13	1001	3500
Sounding	600		19	3501	5000

4.1.3 Acceptance and rejection. - A sample shall be accepted or rejected on requirements listed under group A or under B tests by the appropriate plan given in Table VI. A lot shall be accepted if the sample representating that lot meets the requirements of this specification and it shall be rejected if the sample does not meet these requirements.

Table VI

Acceptance - Rejection Plan

Test Group	Sample Size	Acceptance Number (Defective)	Rejection Number (Defective)
A	6	1	2
	9	2	3
	13	3	4
	19	5	6
B	3	1	2

The sample shall also meet all applicable requirements of 3.10, 3.11, and 3.13.

Test	Reference
Group A Tests	
Weight	3.4, 4.3.3
Flaccid length	3.5, 4.3.4
Neck length	3.6.2, 4.3.5.1
Neck diameter	3.6.3, 4.3.5.2
Group B Tests	
Neck wall thickness	3.6.4, 4.3.5.3
Neck strength	3.6.5, 4.3.5.4
Color	3.7, 3.7.1, 3.7.2, 3.7.3 3.7.4, 3.7.5, 4.3.6

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Group B Tests (Continued)

Shape	3.8, 4.3.7
Bursting dimensions	3.9, 4.3.8
Ozone resistance	3.12, 4.3.11

4.2 Inspection. - The manufacturer shall partially inflate and visually examine each balloon after completion of all manufacturing processes and immediately before packaging. Any leaky balloons, balloons which have been damaged by handling, or balloons which have defects in the film shall be eliminated during this inspection. Balloons which have been dried in a deflated or only partially inflated condition will have an unusually thick film and too short a flaccid length. All such "deflates" shall be detected and eliminated. Balloons which are not properly coated with talc or equivalent material shall either be so coated or eliminated. The neck of the balloon shall be inspected and any balloons which do not conform to the requirements of 3.6.1 and 3.6.6 shall be changed to conform or eliminated. A Government inspector shall observe the manufacturer's inspection as necessary to ascertain that it is properly done and that the balloons are not more than 240 days old when inspected. (See 3.3.)

4.3 Tests.

4.3.1 Equipment and methods. - When required by the proposal request, or contract provisions, the contractor shall furnish any or all of the facilities, equipment, and personnel necessary for making the ground tests of this specification. The equipment employed shall be of a quality acceptable to the Government and adequate in quantity to enable testing to keep up with production. Devices for the measurement of temperature, pressure, weight, elongation, and other physical properties shall be of known and acceptable accuracy. Data taken in the comparison of such devices with approved standards shall be made available to the Government upon request. A Government representative may observe the tests performed by the contractor and ascertain that they are properly accomplished. Flight tests will be made by Government agencies at no expense to the contractor.

4.3.2 Preconditioning. - Balloons tested under this specification shall be preconditioned in accordance with the instructions furnished with the balloon (see 5.4.1.1), where such instructions have been approved by the contracting officer. Unless otherwise specified, dumbbell and patch test specimens shall be placed in boiling water for five minutes and then dried for at least 12 hours before testing. Balloons for inflation or flight tests shall be placed in boiling water for five minutes and inflated or flown soon afterwards except that, if the temperature is below 5°C., the balloons must not be flown until their surface is dry. An alternative method of preconditioning balloons for inflation or flight tests is to place the balloon inside its container in an oven at a temperature of approximately 60°C., for 12 to 24 hours just before testing.

4.3.3 Weight. - All balloons in a sample shall be weighed on a balance which is accurate to one half a gram or less. The weight of a balloon includes the weight of the envelope, the neck, and the talc or equivalent material within and on the balloon, but no packages. If a balloon contains more talc or equivalent material than is necessary to prevent the folds of the film from adhering to each other, the Government may require the excess talc to be emptied out before weighing. (See 3.4.)

4.3.4 Flaccid length. - The flaccid length is the length of the uninflated envelope of the balloon when suspended by the neck. It does not include the length of the neck. The flaccid length of each balloon in a sample shall be measured to the nearest half inch. (See 3.5.)

4.3.5 Neck

4.3.5.1 Neck length. - The neck length is the distance from the point at which the neck becomes of approximately constant cross-sectional area to the free end, except that, if the design is such as to include a collar or enlarged portion between the neck and the envelope, the collar shall be considered as part of the neck and not as part of the envelope. The neck length of each balloon in a sample shall be measured to the nearest half inch. (See 3.6.2.)

4.3.5.2 Neck diameter. - The inside diameter of the neck of each balloon in a sample shall be checked by fitting the neck over rigid tubes or cylinders of circular cross-section. The diameters of these gages shall be as given in Table II. The neck shall fit over the gage of minimum diameter without stretching, but it shall require stretching to fit over the gage of maximum diameter. The

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number of balloons which meet this requirement and the number which do not meet it shall be noted. (See 3.6.3.).

4.3.5.3 Neck wall thickness. - The average thickness of the neck wall shall be measured on three (3) balloons using the method given in Federal Specification ZZ-R-601. It is necessary to cut the neck apart in order to measure its thickness. This test may be performed on ceiling or pilot balloons that have been used for the measurement of bursting dimensions discussed in 4.3.8, pilot balloons that have been used to cut dumbbell specimens for the measurement of breaking strength and ultimate elongation described in 4.3.9, or sounding balloons that have been used for the measurement of breaking strength and ultimate elongation (4.3.9), the low temperature patch test (4.3.10) and the test of neck strength (4.3.5.4). (See 3.6.4.)

4.3.5.4 Neck strength. - This test applies only to sounding balloons. Its purpose is to test the strength of the neck and the region where the neck joins the body of the balloon. It is done in the following manner. A metal tube or cylinder with a smooth outer surface and a circular cross-section two (2) inches in diameter is held with its axis horizontal. The neck and approximately 12 inches of the envelope of the balloon is cut off; the cut end of this section of the balloon is clamped along the bottom of the cylinder, and the section of the balloon is wrapped three-fourths of the way around the cylinder. A 40 pound weight is attached to the free end of the neck and the cylinder is raised slowly. The section of the balloon must support the 40 pound weight for 30 seconds. This test is performed on the three (3) balloons from which patches and dumbbell specimens are cut for the tests of 4.3.9 and 4.3.10 and the number of failures is noted. (See 3.6.5.)

4.3.6 Color. - This paragraph does not apply to uncolored balloons. The balloons tested by the method of 4.3.8 shall be visually examined against a dark background at the volume or diameter specified in Table III for conformance to the requirements of 3.7 and 3.7.1, 3.7.2, 3.7.3, or 3.7.4.

4.3.7 Shape. - This test applies only to ceiling and pilot balloons. The balloons tested by the method of 4.3.8 shall be visually examined at the volume or diameter specified in Table III. If the sphericity appears questionable, a Government representative may, at his discretion, require that the major and minor diameters be measured either by taking photographs of the balloon, by measuring the shadow of the balloon, by using optical methods, or by using calipers or other mechanical means, and the diameter ratio determined. The major diameter is the longest straight line that can be drawn between any two points on the surface of the balloon 45 degrees or more removed from the neck. The minor diameter is the shortest straight line between any two points on the surface of the balloon which passes through the midpoint of the major diameter. The diameter ratio is the ratio of the minor diameter to the major diameter expressed in percent. (See 3.8.)

4.3.8 Bursting dimensions. - Three (3) balloons from each lot of ceiling and pilot balloons shall be suspended by the neck and inflated with air until they burst at a temperature between 5°C and 40°C. Either the volume or the diameter of the balloon at burst shall be measured. The rate of inflation shall be such that either the value of the volume or the value of the diameter given in Table IV would be reached in from 40 to 60 minutes for pilot balloons and in from 30 to 60 minutes for ceiling balloons. (See 3.9.)

4.3.9 Breaking strength and ultimate elongation. - This test does not apply to ceiling balloons. Three (3) dumbbell specimens of the type described in Item D, Figure 2 of Federal Specification ZZ-R-601, having a width of 0.25 ± 0.01 inches and a gage length of 1.0 inch are cut from each of three (3) balloons.

The specimens shall be cut from different regions of the balloon so as to be representative of the entire area of the envelope. After cutting, the specimens shall be carefully inspected for flaws along the edges and any which have such flaws shall be replaced before testing. The nine (9) specimens shall be stretched to failure on a tensile machine at a temperature between 5°C and 40°C, following the method of Federal Specification ZZ-R-601 except that unusually low readings are not thrown out and only the median breaking force and the median ultimate elongation are determined. (See 3.10.)

4.3.10 Low temperature patch test. - This test applies only to sounding balloons. Three (3) circular patches 3 inches in diameter are cut from each of three (3) balloons. The patches should be cut from different regions of the balloon and they shall not be taken from areas immediately adjacent to the neck or other points where the envelope is obviously of greater thickness due to the manufacturing process involved. The patches selected must be as representative as possible of the entire

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envelope. They may be cut from the same balloons used to cut the dumbbell specimens in paragraph 4.3.9. Any patch that is damaged in handling should not be tested, but should be replaced by another from the same balloon. A 1 cm. diameter circle is marked in the center of each patch using ordinary stamp pad ink on a metal die. The patches are placed in the ring shown in Figure 1, clamped to the base shown in Figure 1, and inflated until they burst in a temperature-controlled chamber. The elongation is determined by reading the diameter of the circle through a window in the chamber, using a flexible scale which follows the shape of the patch. The differential pressure between the inside and outside of the patch is read at elongations of 300%, 350%, 400%, 450% and at burst. The inflation is done at a constant rate such that a normal burst will occur in 15 to 20 minutes. The patch is inflated at a temperature of $-40 \pm 2^{\circ}C$, until an elongation of 150% is reached, then the temperature is lowered to $-60 \pm 1^{\circ}C$, and maintained there for the duration of the test. The cooling should be accomplished rapidly enough so that the temperature reaches $-60 \pm 1^{\circ}C$, at least five (5) minutes before the patch bursts. (See 3.11.)

4.3.10.1 Effective elongation. - When three (3) patches are cut from each of three (3) balloons and tested by the method of 4.3.10, the effective elongation is determined in the following manner: The average differential pressures of all nine (9) patches at elongations of 300%, 350%, 400%, 450% and at burst are calculated and a curve of pressure vs. elongation plotted by connecting these points with straight lines. The range in bursting pressure is found for each of the three (3) balloons, then the average of all three (3) ranges is found. The minimum bursting pressure is taken as the average bursting pressure for the nine (9) patches minus 1.18 times this average range. (The factor 1.18 is correct only when three (3) patches per balloon are used). The effective elongation is simply the elongation corresponding to the minimum bursting pressure on the pressure vs. elongation curve.

4.3.11 Ozone resistance. - This test applies only to sounding balloons. One (1) dumbbell specimen of the type referred to in 4.3.9 is cut from each of the three (3) balloons used in the tests described in 4.3.9 and 4.3.10, stretched to 350% elongation and exposed to a concentration of 0.010% to 0.015% ozone for 15 minutes. At this time the number of specimens which have broken is noted. (See 3.12.)

4.3.12 Flight elevation. - This paragraph applies only to sounding balloons. Flight tests shall be conducted by Government agencies. The balloons may be preconditioned by either of the methods given in 4.3.2 unless special instructions are furnished with the balloon (see 5.4.1.1) and such instructions have been approved by the contracting officer. The balloons shall be inflated with helium under ambient conditions of temperature and pressure so that the free lift is 1200 to 1500 grams when carrying a radiosonde train weighing 1200 to 1700 grams. The flights shall be made in the continental United States at any season of the year. Balloons shall be flown during the day and at night in the quantities shown in Table VII. The median day flight elevation and the median night flight elevation shall be determined. If the bursting elevation on a flight is not obtained because of instrument failure or weak signal, the flight will not be used in determining the median unless the balloon has already reached the required elevation of 23.5 kilometers. A flight terminated because of a premature burst which was obviously due to mishandling during preconditioning, inflation, or launching will not be used in determining the median. A flight terminated because of a premature burst or a leak in the balloon which was not obviously caused by mishandling during preconditioning, inflation, or launching but may be due to a flaw in the balloon will be used in determining the median. When the number of flights, usable for determining median altitudes attained, under the foregoing rules is less than the minimum numbers shown in Table VII, an additional quantity of sample balloons from the same production lot will be obtained and a sufficient number flown to insure that the median is based on the minimum numbers given in Table VII for the production lot in question. (See 3.13.)

Table VII

Flight Test Plan - Sounding Balloons Only

Lot Size	Total Sample Size	Flights Required		Minimum Number of Flights Required in Determining Median	
		Day	Night	Day	Night
500 to 1000	9	3	3	2	2
1001 to 3500	13	5	5	3	3
3501 to 5000	19	8	8	5	5

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5. PREPARATION FOR DELIVERY

5.1 Packaging. - Balloons shall be furnished in packages constructed of cardboard or similar material which will not injure the balloons and so designed as to provide adequate protection. The seams and openings of the packages shall be sealed so as to exclude light and to be practically dustproof. The package shall be designed so that any portion of a balloon extending over the edge will not be easily damaged when the package is closed. Each sounding balloon shall be placed in a paper bag before being packaged.

5.2 Packing for Domestic Shipment. - Unless otherwise specified, the balloons in the packages described in paragraph 5.1 shall be placed in corrugated or solid fiberboard boxes having a minimum Mullen test strength of 200 pounds. All flaps shall be closed by stitching, taping, sealing with a suitable adhesive or by a combination of these methods.

5.3 Packing for Export Shipment. - When export shipment is required by the contract, the containers specified in paragraph 5.1, above, shall be prepared for shipment as specified in the military specifications listed in paragraph 2.2. Options as to types, grades, classes and methods provided for in these specifications, pertaining to packages, containers, materials and procedures, shall be exercised by the contractor so as to insure adequate protection and economical handling of the product under conditions of rigorous military shipment, storage and reshipment.

5.4 Marking

5.4.1 Packages. - Packages shall be marked with the function, nominal weight, color, manufacturer's lot number, month and year of manufacture, quantity of the balloon contained, the contract or order number and the name of the contractor.

5.4.1.1 Preconditioning instructions. - Balloons which require special preconditioning procedures before inflation shall have printed on the package, or on a label attached thereto, detailed instructions for such preconditioning. The text of the instructions shall be submitted to and approved by the contracting officer prior to printing.

5.4.2 Shipping containers. - Unless otherwise specified, the outside of all shipping containers shall be marked with the function, nominal weight, color, manufacturer's lot number, month and year of manufacture, quantity of the balloon contained, the contract or order number and the name of the contractor.

5.5 Quantities. - Unless otherwise specified, the number of balloons in a package and the number of packages in a shipping container shall be as shown in Table VIII.

Table VIII

Quantities

Function	Nominal Weight, Grams	No. of Balloons per Package	No. of Packages per Shipping Container
Ceiling	10	50	32
Pilot	30	20	32
Pilot	100	10	32
Sounding	600	1	32

6. NOTES

6.1 Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

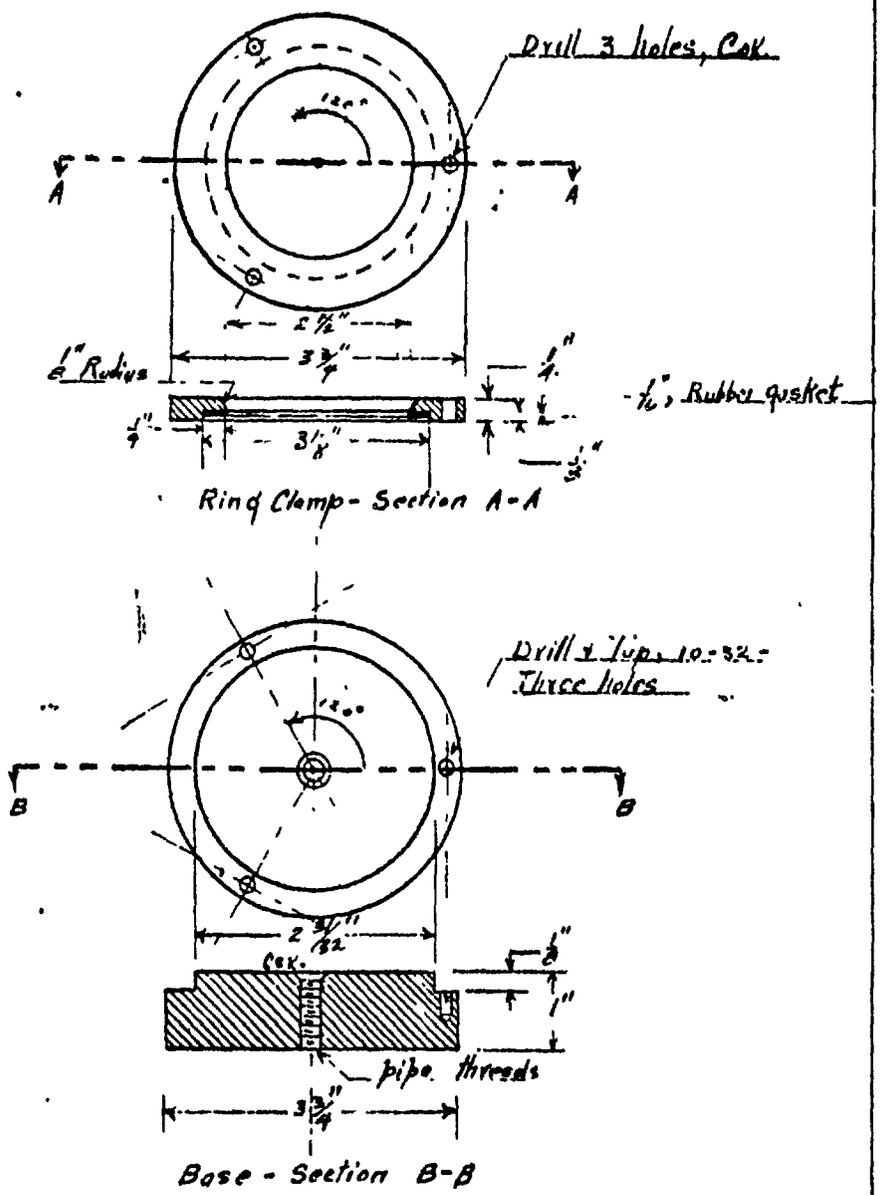
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6.2 Service conditions.-The balloons furnished under this specification are used for meteorological purposes. Normally ceiling and pilot balloons carry no load and are followed by visual means, while sounding balloons transport weather instruments into the stratosphere and are followed by means other than visual. The balloons are used in all climates and in all weather conditions. They are inflated with helium or hydrogen and, when released, travel upwards at varying rates depending upon their size and bouyancy. The same balloon may be exposed to temperatures as high as 50°C and as low as -70°C. The atmospheric pressure on the ground may be as much as 1050 millibars and sounding balloons may travel to regions where the pressure is as low as 10 millibars. The wind speed at the time of release may be 30 knots or more. During launching a balloon is held by the neck; the neck of a sounding balloon may occasionally be required to withstand a pull of as much as 40 pounds. During the flight the humidity may range from 100% to 0% and all types of precipitation and icing may be encountered. Sounding balloons may reach altitudes at which otone, ultra violet radiation and cosmic radiations are much more prevalent than at the surface of the earth.

6.3 Storage. - The manufacturer shall guarantee that the balloon material contains sufficient anti-oxidant and anti-freeze substances so that the balloons in their original packages may be stored for as long as eighteen (18) months under normal conditions and then, after preconditioning, flown successfully.

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FIGURE-1



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